

WASP-104b

R-2407

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-104_190224 · created 2026-08-21T01:09:57+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458538.9266 +/- 0.0021 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.152 +/- 0.042
Transit depth (Rp/Rs)^2	2.3 +/- 1.27 [%]
Orbital Inclination (inc)	81.42 +/- 2.88
Airmass coefficient 1 (a1)	0.919 +/- 0.029
Airmass coefficient 2 (a2)	0.054 +/- 0.015
Scatter in the residuals of the lightcurve fit is	3.16 %
Best Comparison Star	#2 - [139, 451]
Optimal Aperture	4.02
Optimal Annulus	32.13
Transit Duration (day)	0.047 +/- 0.034

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.152 ± 0.042 vs 0.1214 (deviation 0.5σ)
Duration fit vs published	0.047 d vs 0.0742 d (deviation 0.54σ)
Mid-time O–C	-4.7 min
Depth SNR	2.5
Residual scatter	3.16 %

Light curve

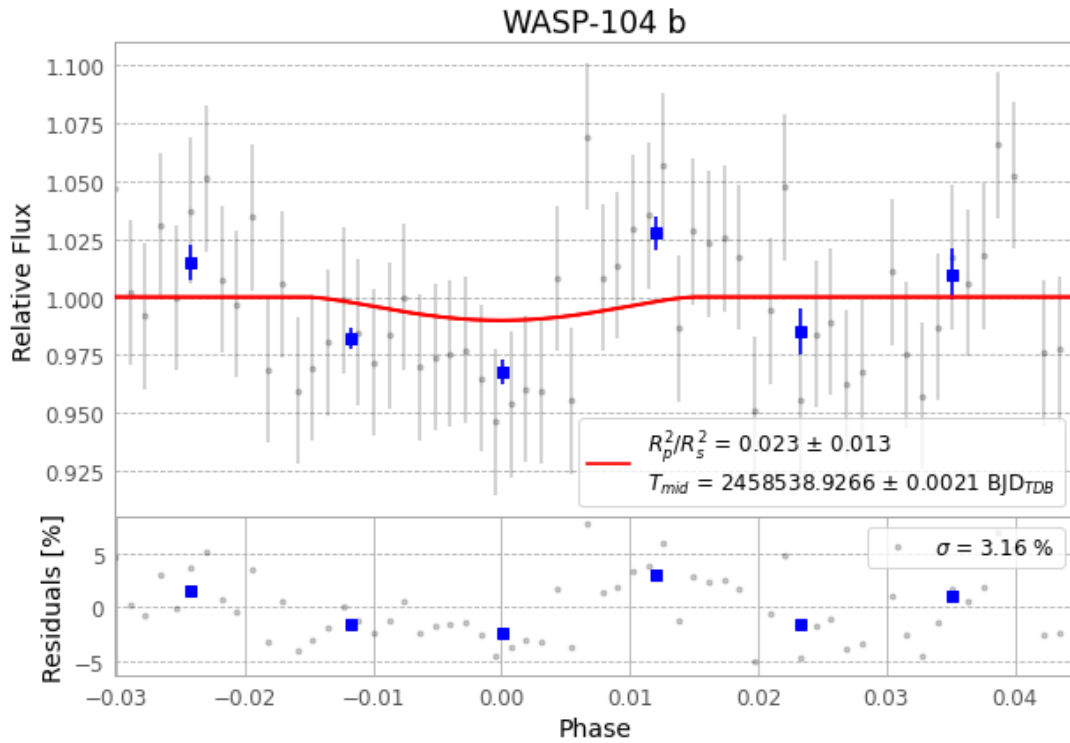


Figure 1 — FinalLightCurve_WASP-104 b_2019-02-24.png (SHA-256 235b6375e781cd19...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [363, 235], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 bd76c57765d3bdca...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

64 FITS frames (42 MB), WASP-104190224084813.FITS → WASP-104190224115712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-104-190224.log (63db7faf572f...), FinalLightCurve_WASP-104 b_2019-02-24.png (235b6375e781...), FinalLightCurve_WASP-104 b_2019-02-24.csv (9c87e0704d7b...)

Compute resources

Wall time 297.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-21 01:09Z.